

Amendment No. 1 (05/2026)

to

AIS-213 : Procedure for the Measurement of Emissions and Fuel Consumption with Air-Conditioning System in Operation for M1 Category Vehicles.

1.0 Title of standard

Substitute following text for existing text:

PROCEDURE FOR THE MEASUREMENT OF EMISSIONS, FUEL CONSUMPTION & RANGE FOR M1 CATEGORY VEHICLES WITH AIR-CONDITIONING SYSTEM IN OPERATION

2.0 Clause No 1.0

Substitute following text for existing text:

This standard describes the procedure for M1 category vehicles with Air-conditioning system in operation with regard to

- i) Measurement of Emissions and fuel consumption for ICE vehicles and
- ii) Measurement of Range for Electric Power Train Vehicles (PEV and PHEV)

3.0 Clause No 2.1

Substitute following text for existing text:

For the purposes of this Chapter, definitions specified in AIS 137 (Part 3) & AIS-049 (Rev. 1): 2016 shall apply.

4.0 Clause No 3.1

Substitute following text for existing text:

Vehicle selected for Emission Certification/ Range Certification as a worst case shall be tested with Air-conditioning System ON as per the procedure stated here.

5.0 Clause No 5.2

Substitute following text for existing text:

Sampling System (Applicable to ICE vehicles)

6.0 Clause No 5.2

Substitute following text for existing text:

Analytical Equipment (Applicable to ICE vehicles)

7.0 Clause No 5.5

Substitute following text for existing text:

Gases (Applicable to ICE vehicles)

8.0 Clause No 9.0

Substitute following text for existing text:

Preconditioning (Applicable to ICE vehicles)

9.0 Clause No 11.1

Add following text after the clause:

(Applicable to ICE vehicles)

10.0 Clause No 11.2

Add following text after the clause:

(Applicable to ICE vehicles)

11.0 Clause No 11.3

Add following text after the clause:

(Applicable to ICE and PEV/PHEV vehicles)

12.0 Clause No 12.0

Substitute following text for existing text:

Calculations (Applicable to ICE vehicles)

13.0 Add clause no 14.0 after clause no 13.0

14.0 Provision for PEV and PHEV Vehicles

Range Test shall be carried out as per the provision of AIS-040(Rev.1):2015 amended from time to time

Test Cell Conditions as per Clause 7.0 & Test Preparation as per clause no 8.0 shall be followed.

The test vehicle (and consequently the air conditioning system) and cooling fan(s) are turned off and the vehicle is allowed to soak in the ambient conditions of Clause 7.0 of this section for 10 ±1 minute after end of End of charge criteria (as per AIS-039(Rev.1):2015 & AIS-040(Rev.1):2015 as applicable) & before commencement of the test cycle.

14.0 Clause no 8.4

Substitute following text for existing text:

8.4 To take account of the difference in air density at 35°C when compared to the air density at 20°C, the chassis dynamometer shall be set in accordance with the Type Approval Type I Test as per AIS 137 Part 3 with the exception that second order road load coefficient at 35°C ($F_{2_35^{\circ}\text{C}}$) from the following equation shall be used as the target coefficient C_t .

$$F_{2_35^{\circ}\text{C}} = F_2 \times (T_{\text{ref}} + 273)/(T_{\text{reg}} + 273)$$

where:

F_2 is the second order road load coefficient, at reference conditions, $N/(km/h)^2$;

T_{ref} is the road load reference temperature i.e. $20^\circ C$;

T_{reg} is the regional temperature i.e. $35^\circ C$.

In the case that a valid chassis dynamometer setting of the $23^\circ C$ test is available, the second order chassis dynamometer coefficient of C_d shall be adapted in accordance with the following equation:

$$C_{d_{35^\circ C}} = C_d + (F_{2_{35^\circ C}} - F_2)$$

15.0 Clause no 12.1

Add following text at the end of clause

Alternatively, Fuel consumption (l/100km) can be calculated as follows:

- $FC_{Petrol} = 0.04217 * CO_2$
- $FC_{Diesel} = 0.03776 * CO_2$
- $FC_{LPG} = 0.06150 * CO_2$
- $FC_{CNG} = 0.03647 * CO_2$

PRINTED BY

THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA

P. B. NO. 832, PUNE 411 004

ON BEHALF OF

AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER

CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY

MINISTRY OF ROAD TRANSPORT & HIGHWAYS

GOVERNMENT OF INDIA

7th May 2026

AUTOMOTIVE INDUSTRY STANDARD

**Procedure for the Measurement of Emissions
and Fuel Consumption with Air-Conditioning
System in Operation for M1 Category
Vehicles.**

PRINTED BY
THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA
P.B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

November 2024

Introduction

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CMVR-TSC). After approval, The Automotive Research Association of India, (ARAI), Pune, being the Secretariat of the AIS Committee, will publish this standard. For better dissemination of this information, ARAI may publish this document on their Web site.

The intention of this procedure is to establish uniform provisions of motor vehicles of Category M1 with regard to measurement of fuel consumption of the Vehicles with air conditioning in operation.

During the 71st Automotive Industry Standard (AIS) Committee meeting, it was decided frame a procedure for fuel economy evaluation for passenger cars with air conditioning system operational based on a procedure followed by EPA, USA.

There is no test procedure available in EU or Japan or UNECE for evaluation of fuel economy of light duty passenger vehicles in a laboratory setting with AC on condition. USEPA has established SFTP SC03 test cycle and procedure for evaluating emissions and fuel economy of light duty vehicles in laboratory with AC on condition. Since 2008, the fuel economy results from SC03 tests are also used as one of the components in its composite fuel economy labelling program. Hence, the SFTP SC03 test procedure was considered for adoption in AIS.

While preparing the procedure for India, some deviations were taken from the above said procedure in order to address India specific test conditions as defined in AIS 137 Part 3, the available test facilities within India and the solar conditions prevailing in India.

This procedure simulates driving behaviour with the air conditioning system operating with engine startups and vehicle operation on a chassis dynamometer through Modified Indian Driving Cycle (MIDC). A proportional part of the diluted exhaust is collected continuously in bags for subsequent analysis, using a constant volume sampler. The testing sequence consists of a preconditioning cycle, a 10-minute soak with the engine turned off, and the MIDC with exhaust emissions measurement.

The deviations introduced in the regulation with respect to SC03 Procedure are tyre pressure setting, test cell conditions, soaking duration and air conditioning system settings. Test cell conditions also include Solar radiation simulation for which the considerable assistance is taken for the specifications from SC03 Procedure. However, for the absolute solar radiation intensity setting, data was collected and analysed for 12 major cities as well as over entire India to arrive at the solar intensity setting. In addition, other procedural requirements such equipment, calibration, fuel, coast down, test cycle, exhaust gas sampling, bag analysis and calculations are to be referred from AIS 137 Part 3.

The vehicle identified as a worst-case vehicle for Type-1 Type approval test will be subjected to this testing.

Composition of the Panel and Automotive Industry Standards Committee (AISC) responsible for preparation and approval of this standard are given in Annexure A & B respectively.

CONTENTS

ClauseNo.	Details	Page No.
1.0	Scope	1/7
2.0	Definitions	1/7
3.0	Requirements for Test Vehicle	1/7
4.0	Requirements for Test Fuel	1/7
5.0	Requirements for Test Equipment	1/7
6.0	Vehicle Installation	3/7
7.0	Test Cell Conditions	3/7
8.0	Test Preparation	3/7
9.0	Preconditioning requirements	4/7
10.0	Soaking requirements	4/7
11.0	Test Procedure	4/7
12.0	Calculations	5/7
13.0	Provision for Hybrid Vehicles	5/7
Annex A	AISC Panel Composition.	6/7
Annex B	Automotive Industry Standards Committee Composition.	7/7

Procedure for the Measurement of Emissions and Fuel Consumption with Air-Conditioning System in Operation for M1 Category Vehicles.

1.0 SCOPE

This standard describes the procedure for the measurement of emissions and fuel consumption with Air-conditioning system in operation for M1 category vehicles.

2.0 DEFINITIONS

2.1 For the purposes of this Chapter, definitions specified in AIS 137 (Part 3) shall apply.

3.0 REQUIREMENTS FOR TEST VEHICLE

3.1 Vehicle selected for Emission Certification as a worst case shall be tested with Air-conditioning System ON as per the procedure stated here.

3.2 Air conditioning system of the vehicle shall be fully operational as per the design specifications.

3.3 The provisions of Chapter 3 of AIS 137 (Part 3) apply.

4.0 REQUIREMENTS FOR TEST FUEL

4.1 The provisions of Clause 2.3 of Chapter 3 AIS 137 (Part 3) apply.

5.0 REQUIREMENTS FOR TEST EQUIPMENT

Equipment required and specifications are equivalent to the requirements for the Type I Test as specified in AIS 137 (Part 3), with appendices, if specific requirements for this test are not prescribed.

5.1 Chassis Dynamometer and Vehicle Cooling Fan

5.1.1 The provisions of Chapter 3, 4 and 5 of AIS 137 (Part 3) apply with the following exception:

The minimum airflow nozzle discharge area must be equal or exceed the vehicle frontal inlet area. Optimum discharge area is 1.7 m². However, other sizes can be used.

5.2 Sampling System

5.2.1 The provisions of Chapter 3, 4 7 and 8 of AIS 137 (Part 3) apply

5.3 Analytical Equipment

5.3.1 The provisions of Chapter 7 and 8 of AIS 137 (Part 3) apply

5.4 For calibrations of the analytical equipment the provisions of AIS 137 (Part 3) apply.

5.5 Gases

5.5.1 The provisions of Clause 3 Chapter 7 of AIS 137 (Part 3) apply, where they are relevant.

5.6 Solar Simulation System

5.6.1 Solar heating shall be simulated as follows:

Metal halide lamp, a sodium lamp, or a quartz halogen lamp with dichroic mirrors as a radiant energy emitter shall be used.

5.6.2 It is recommended to achieve radiant heating with spectral distribution characteristics as described in the following table:

5.6.3 Recommended Spectral Distribution

Band width (nm)	Percent of total spectrum	
	Lower limit (%)	Upper limit (%)
<320 ^a		0
320–400	0	7
400–780	45	55
>780	35	53
^a Note that the UV region between 280 and 320 nm may be filtered.		

5.6.4 Determine radiant energy intensity experienced by the vehicle as the average value between two measurements along the vehicle's centreline, one at the base of the windshield and the other at the bottom of the rear window (or equivalent location for vehicles without a rear window). This value must be (780 ± 45) W/ m². Instruments for measuring radiant energy intensity must meet the following minimum specifications:

5.6.4.1 Sensitivity of 9 microvolts per W/m².

5.6.4.2 Response time of 5 seconds. For purposes of this requirement, “response time” means the time for the instrument to reach 95 percent of its equilibrium response after a step change in radiant intensity.

5.6.4.3 Cosine response error of no more than $\pm 1\%$ for 0–70 degree zenith angles. The cosine response error is the percentage difference between the intensity measured at a given angle and a reference value, where the reference value is the intensity predicted from the zero-degree intensity and the cosine of the incident angle.

- 5.6.4.4 When comparing measured values for radiant energy to reference values, each measured value over the full range of measurement may not deviate from the corresponding reference value by more than $\pm 0.5\%$ of the analyzer range's maximum value.
- 5.6.5 Check the uniformity of radiant energy intensity at least every 500 hours of emitter usage or every 12 months, whichever is sooner, and after any major modifications affecting the solar simulation. Determine uniformity by measuring radiant energy intensity using instruments that meet the specifications described in 5.6.4 at each point of a 0.5 m grid over the vehicle's full footprint, including the edges of the footprint, at an elevation 1 m above the floor. Measured values of radiant energy intensity must be between (722 and 978) W/m^2 at all points.

6.0 VEHICLE INSTALLATION

- 6.1 Inertia Class and Road Load Equation shall be same as used during Type Approval Type I Test as per Chapter 3 AIS 137 (Part 3).
- 6.2 Vehicle installation shall be same as Type Approval Type I Test as per Chapter 3 AIS 137 (Part 3).
- 6.3 Tyre Pressure shall be same as used during Type Approval Type I Test as per AIS 137 (Part 3) and shall be recorded.

7.0 TEST CELL CONDITIONS

- 7.1 During the preconditioning, soaking and test, the test cell conditions shall be as per Clauses 7.2, 7.3 and 7.4. Measure and record the temperature and humidity in the test cell with the frequency of at least 1 Hz.
- 7.2 Temperature shall be set and maintained within 35 ± 3 °C on average.
- 7.3 Relative Humidity shall be set and maintained within 45 ± 8 % RH on average.
- 7.4 Solar heat load intensity of 780 ± 45 W/m^2

8.0 TEST PREPARATION

- 8.1 All vehicle windows shall be closed during the tests.
- (i) Install the vehicle on the chassis dynamometer and ensure the correct tyre pressure in all tyres. Make connections of exhaust system.
 - (ii) Turn on the solar heating system.
- 8.2 Battery may be charged before the preconditioning cycle.
- 8.3 Air conditioning system controls shall be set as follows:
- (i) A/C mode setting at Maximum.

- (ii) Airflow setting at Recirculate, if so equipped.
- (iii) Fan setting at Highest setting.
- (iv) A/C Temperature setting at full cool (for automatic systems set at 24 °C).
- (v) For vehicles fitted with dual zone/three zone/multiple zone AC systems, above setting shall be done for the Driver and Co-Driver compartments.
- (vi) Air conditioning controls should be placed in the “on” position prior to vehicle starting so that the air conditioning system is active whenever the engine is running.

8.4 Coastdown Derivation for Dyno set shall be same as used during Type Approval Type I Test as per AIS 137 Part 3

8.5 Provisions of Clause 6.0 of Chapter 3 of AIS 137 Part 3 apply with the exception to clauses 8.1 to 8.3.

9.0 PRECONDITIONING

9.1 Start the vehicle (with air conditioning system on) and conduct a preconditioning cycle as used during Type Approval Type I Test as per Clause 6.10 of Chapter 3 of AIS 137 Part 3.

10.0 SOAKING

10.1 Following the preconditioning cycle, the test vehicle (and consequently the air conditioning system) and cooling fan(s) are turned off and the vehicle is allowed to soak in the ambient conditions of Clause 7.0 of this section for 10 ±1 minute.

11.0 TEST PROCEDURE

11.1 Start the engine (with air conditioning system also running). All the other provisions of Clause 6.11 of Chapter 3 AIS 137 Part 3 apply in respect of starting the engine, carrying out the test and taking the emission samples.

11.2 For the Bag analysis of the sampled emissions the provisions of Clause 6.12.3 of Chapter 3 AIS 137 Part 3 apply with following deviation.

Pollutants to be measured and Fuel Consumption to be calculated:

For vehicles with a compression ignition engine: THC (heated continuous recording or Bag measurement in case heated continuous recording is not possible), CO, CO₂, NO_x. PM and PN measurement are optional.

For vehicles with a positive ignition engine: THC, CO, CO₂, NO_x. PM and PN measurement are optional for vehicles with direct injection engine.

11.3 During the test, temperature inside the vehicle cabin shall be recorded with frequency of min 1 hz at driver and co-driver location, preferably near nose level.

12.0 CALCULATIONS

- 12.1 For the calculations of the mass emissions the provisions of Clause 6.13 of Chapter 3 AIS 137 Part 3 apply. The results shall be multiplied with Regenerating Factor (Ki) and Deterioration Factor (DF).

For the calculations of FUEL CONSUMPTION, the provisions of Clause 7.0 Chapter 3 AIS 137 Part 3 apply.

13.0 PROVISION FOR HYBRID VEHICLES

- 13.1 For NOVC Hybrid Vehicles - Test shall be carried out as per above procedure and with battery charge balance measurement. Kfuel and KCO2 factors - Same as used during Type Approval Type I Test.

For OVC Hybrid Vehicles - Test shall be carried out as per above procedure in Condition A and Condition B as defined in Chapter 17 and 19A of AIS 137 Part 3.

ANNEX A
(See Introduction)

Composition of AISC Panel*

Name	Organization
Convener	
Dr. P.G. Bhat	The Automotive Research Association of India (ARAI)
Members	
Mrs. Sheetal S. Bakle	The Automotive Research Association of India (ARAI)
Mr. Prashant N. Pawar	The Automotive Research Association of India (ARAI)
Mr. Vikram Tandon	The Automotive Research Association of India (ARAI)
Mr. Moqtik Bawase	The Automotive Research Association of India (ARAI)
Mr. Kamalesh Patil	The Automotive Research Association of India (ARAI)
Mr. Shantanu Tijare	The Automotive Research Association of India (ARAI)
Mr. Yogesh Sathe	The Automotive Research Association of India (ARAI)
Mr. Vikas Sadan	ICAT
Mr. Deepak Joshi	ICAT
Mr. Sitikantha Padhy	ICAT
Mr. V.M. Dhanasekar	GARC
Mr. P. K Banerjee	SIAM
Dr. Sandeep Garg	SIAM
Mr. P. Gowrishankar	SIAM (Tata Motors Ltd.)
Mr. Gururaj Ravi	SIAM (Maruti Suzuki India Ltd.)
Mr. Rajendra Khile	SIAM (Renault Nissan India Pvt. Ltd.)
Mr. Gnanasekaran K	SIAM (Renault Nissan India Pvt. Ltd.)
Mr. Sanjay Tank	ACMA

ANNEX B
(See Introduction)

COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

Chairperson	
Dr. Reji Mathai	Director, The Automotive Research Association of India
Members	Representing
Representative from	Ministry of Road Transport and Highways
Representative from	Ministry of Heavy Industries
Representative from	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises
Shri Shrikant R. Marathe	Former Chairman, AISC
Head-TED	Bureau of Indian Standards
Director	Central Institute of Road Transport
Director	Global Automotive Research Centre
Director	International Centre for Automotive Technology
Director	Indian Institute of Petroleum
Director	National Automotive Test Tracks
Director	Vehicles Research and Development Establishment
Director	Indian Rubber Manufacturers Research Association
Representatives from	Society of Indian Automobile Manufacturers
Representatives from	Tractor and Mechanization Association
Representatives from	Automotive Components Manufacturers Association of India
Representative from	Indian Construction Equipment Manufacturers' Association
Member Secretary	
Shri Vikram Tandon	The Automotive Research Association of India

* At the time of approval of this Automotive Industry Standard (AIS)